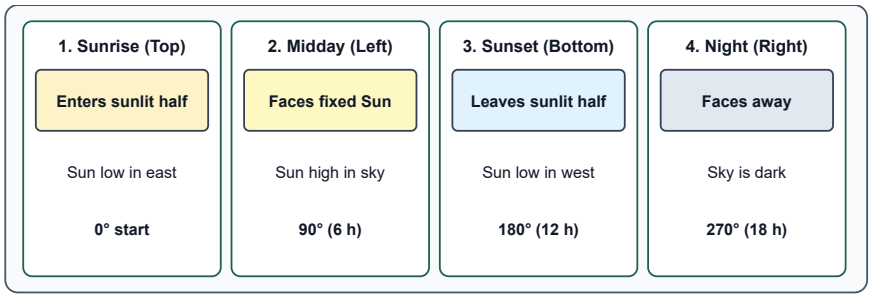
Reference sheets

Reference A · One marked place rotating through light and dark (North-Pole view)

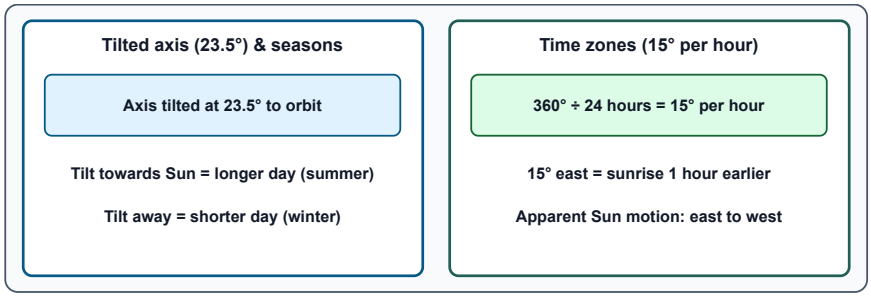


Four positions of a marked place in the North-Pole view with sunlight from the left: 1. Sunrise, 2. Midday, 3. Sunset, 4. Night.

Reference only. Do not write on this sheet. Keep the Sun fixed on the left and follow the red marker anticlockwise (1 → 2 → 3 → 4).

Position in North-Pole view	Time of day	Space viewpoint clue (what the marker does)	Observer viewpoint clue (how the sky looks)
Position 1 · Top boundary	Sunrise	Turns anticlockwise from the unlit half into the sunlit left half	Sun appears low in the eastern sky
Position 2 · Left sunlit side	Middle of the day	Faces directly towards the fixed Sun on the left	Sun appears high across the sky
Position 3 · Bottom boundary	Sunset	Turns anticlockwise out of the sunlit half into the unlit right half	Sun appears low in the western sky
Position 4 · Right unlit side	Night	Faces away from the fixed Sun inside the shaded right half	Sky is dark; the Sun is below the horizon

Reference B · Key Stage 3: tilted axis (23.5°), day length, seasons, and time zones



Key Stage 3 model: Earth's 23.5° tilted axis causes changing day length and seasons, while 360° rotation in 24 hours (15° per hour) causes time zones.

Key Stage 3 concept	Space mechanism (actual cause)	Observable effect on Earth
Apparent Sun motion	Earth rotates west to east on its axis once every 24 hours	The observer turns with Earth, so the Sun appears to move from east to west across the sky
Time zones (15° per hour)	Earth rotates 360° in 24 hours, turning 15° of longitude each hour (360° ÷ 24 = 15°)	Global time zones shift by 1 hour for every 15° of longitude east or west
Tilted axis & length of day	Earth's axis is tilted at 23.5° relative to its orbit around the Sun	Daylight is longer than 12 hours when our hemisphere tilts towards the Sun and shorter when it tilts away
Seasons	Earth orbits the Sun once a year while keeping its 23.5° axial tilt pointed in the same direction	Hemisphere tilted towards the Sun has summer (longer days, higher Sun); opposite hemisphere has winter (shorter days, lower Sun)

Made by Matt

Tier pack · Independent

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Use the arrival picture cue. Which marker is in darkness, and how does the drawing show this?

Help: Check which half of Earth is shaded and facing away from the light source.

2. The Moon stays spherical all month. Why does the lit shape we see from Earth change during a month?

Help: Think about how our view of the sunlit half changes as the Moon orbits Earth.

3. What is the difference between Earth's daily rotation and Earth's yearly orbit?

Help: Compare spinning on an imaginary pole-to-pole line with travelling around the Sun.

4. Why must we keep the model lamp still while turning the globe to investigate day and night?

Help: Think about which body actually turns to cause the 24-hour cycle.

You do tasks

1. Use Reference A. Track one marked place through Positions 1, 2, 3, and 4 in the North-Pole view. Name the time of day at each position and state the evidence clue.

Staff I ☐ S ☐ C ☐

Help: Keep the Sun fixed on the left and follow the anticlockwise rotation arrow from Position 1 (top) through Positions 2, 3, and 4.

2. Explain why day turns to night using the words axis and rotation, then explain why the Sun appears to move across our sky from east to west. **Staff I ☐ S ☐ C ☐**

Help: Link Earth's west-to-east rotation on its axis to what an observer on Earth sees.

3. Use Reference B to explain how Earth's 24-hour rotation causes time zones and why day length changes through the year. **Staff I ☐ S ☐ C ☐**

Help: Check the 15° per hour row and the 23.5° tilted axis row on Reference B.

Optional participation-loop prompt: Share one explanation with an adult, hear one feedback step, and respond below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. Explain why Place X on the unlit side of Earth has night and how it reaches sunrise and middle of the day.

2. Why does the Sun appear to move across our sky from east to west each day?

3. How does Earth's 23.5° tilted axis change the length of day through the year?

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Independent

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt